

H-60 Improvement Plan

Plain-English version — updated September 6, 2026 — Companion to docs/IMPROVEMENT_PLAN.md (the technical version, same folder)

The short version: the H-60 mod works, but under the hood it has grown messy — duplicate files, conflicting copies of the same settings, a few genuine bugs, an outdated build process, and no automatic checks to stop new mistakes. The two reference mods (the AH-64D Apache and the MH-47G Chinook) solved all of these problems in almost exactly the same way as each other. This plan copies their homework, in seven steps, ordered so each step makes the next one safer. **Status as of September 6, 2026: Steps 0 and 1 are done** — the bug fixes shipped in Stable 0.7.9, and the "one master blueprint" reorganization is merged and awaiting its group test.

What was studied

A full read-through of three codebases: the AH-64D Apache project (the gold standard for project setup — build tools, automated checks, documentation), the PGS MH-47G Chinook (the gold standard for simulating aircraft systems — engines, electrics, warnings), and the H-60 itself (a top-to-bottom audit looking for bugs, clutter, and inconsistency). A bonus discovery: the Chinook is built on the same Hatchet cockpit-interaction framework the H-60 uses, and its comments even credit the H-60 for several tricks. So these patterns are proven to work with this exact setup.

The big idea the reference mods share

Both reference mods, built by different teams, landed on the same rules:

- **One master blueprint.** The helicopter's base definition (its "template") lives in exactly one place. Every other package only adds to it — it never rewrites it. The H-60 previously had four different files all claiming to define the master template, and the game just picked whichever loaded last. That was the root cause of things mysteriously breaking when unrelated parts changed. (*Fixed in Step 1 — see below.*)
- **Numbers live in settings files, not buried in code.** Engine power tables, weights, warning thresholds — kept as data the game can reload live, so tuning doesn't require rebuilding the mod each time.
- **Each helicopter keeps its own scorecard.** Information about a specific aircraft (is engine 1 running? how much does it weigh?) is stored on that aircraft. The H-60 sometimes stores it in shared global variables — which means two Blackhawks in the same mission overwrite each other's data.
- **A robot checks every change.** Both reference projects (especially the Apache) have automated checks that reject badly named files, broken references, and missing translations before a human ever sees them. The H-60's version of this was switched off — the checking step in the automation is literally commented out, and the checker scripts it refers to don't exist.

What was actually wrong with the H-60

1. Real bugs, live right now. The rescue hoist's "raise" function accidentally calls the "lower" function when someone else's machine owns the aircraft (a copy-paste slip). It also targets the wrong helicopter on dedicated servers. A lookup table for the helicopter's paint/parts slots is off by one — five entries point at the wrong slot. The engine startup code initializes engine 1's power twice and engine 2's never, quietly re-registers the same event listener every time you

board (they pile up), and uses shared variables so multiple Blackhawks corrupt each other. Debug leftovers ship to players: one function prints "SETTING UP" chat spam to everyone nearby, another writes to the log every frame.

2. Clutter. About 45 files are dead weight: empty files, abandoned half-finished rewrites, and three different copies of the same important settings file that have drifted apart (108 lines now differ between copies — nobody knows which is "the real one"). There's also an entire hidden Pong game living inside the cockpit-display package.

3. No consistency. The mod has three different styles of registering script functions, four different styles of registering mod options, and almost no translations (248 pieces of English text are hardcoded; only 5 use the translation system). One in-repo package (ace_viv) does everything the right way — the plan is basically "make everything look like that one."

4. An outdated, partly broken build process. The tool that packages the mod is Windows-only, uses a trick that silently skips a whole optimization step (it points the optimizer at a fake program), and the version number is written in four different places that all disagree. Both reference mods use a modern tool called HEMTT that fixes all of this and stamps the version automatically.

The plan, step by step

STEP 0 · SMALL & IMMEDIATE

Fix the real bugs **DONE — shipped in 0.7.9**

Fix the twelve concrete bugs found (hoist, engine startup, off-by-one table, chat spam, plus two confirmed from the public bug tracker: the laser-code error that spams server logs, and the flight-director keybinds that do nothing because they ask for a cockpit knob by the wrong name). No redesign — just corrections. Test with two Blackhawks at once on a dedicated server, since that's where most of these bite. *Shipped as Stable 0.7.9 on September 4, 2026.*

STEP 1 · THE FOUNDATION

One master blueprint **DONE — September 2026**

Merge the competing definitions of the helicopter template into one place, with the others only adding their piece on top. This is the earlier "Snow game plan," but corrected — review found it missed a fifth duplicate (the HH-60 variant), described sound-package content that doesn't exist, and would have silently deleted the minigun cockpit controls and paint schemes if followed word-for-word. Before-and-after snapshots of every helicopter's resolved settings prove nothing changed by accident.

How it actually landed (September 2026): the single home ended up being the *settings* package (uh60_config), not the big model package — and that's better. A quick experiment settled an old assumption: the game doesn't care which package loads first when one definition refers to another by name, so the definitions could move straight to the settings package. The big UH60 package now carries only the art — 3-D models, textures, sounds — which means settings changes no longer force players to re-download a huge file. Along the way the Air Force bird (HH-60) got its own package, and every package now honestly declares what it depends on. Every step was proven with the before/after snapshots: values identical, nothing changed by accident.

STEP 2 · SPRING CLEANING

Delete the clutter

Remove the ~45 dead files, merge the triplicate settings files into one, and fix 12 file references that only work because Windows ignores capitalization ("CfgVehicles" vs "cfgVehicles"). That capitalization fix is required before Step 3 — the new build tool runs on Linux, which does care.

STEP 3 · MODERN TOOLS

New build system + automatic checks

Switch to HEMTT, copying the Apache project's setup: the version number lives in one file and gets stamped everywhere automatically; a one-command launcher starts the game with the right mods for testing; and every change pushed to GitHub gets automatically checked for naming mistakes, broken references, and missing files. Someone on the team already started this migration once (there's an abandoned branch) — this finishes the job.

STEP 4 · ONE WAY OF DOING THINGS

Consistency pass

Pick one style for everything — function registration, mod options, translations, naming — and convert packages to it gradually as they're touched (not in one giant risky rewrite). Also fix the dependency declarations so the game loads packages in a guaranteed order: previously many packages called each other without declaring it, and — backwards — the base helicopter package depended on five of its own add-ons. *(That backwards part is already fixed: it fell out of Step 1's reorganization in September 2026.)*

STEP 5 · THE PAYOFF

Better systems, borrowed from the Chinook

- **Engine & electrics:** the H-60 currently ships two generations of engine code side by side. Finish the new one, delete the old, and structure it the Chinook way — one orderly pass every second (battery → APU → hydraulics → fuel → engines), where flipping any switch mid-start correctly aborts the start. No more wedged states.
- **Warnings:** rebuild the caution/warning system as a simple checklist table (severity, message, condition) that the master-caution light and displays read from — every warning tied to real damage or real system state. Newly added here: a proper advisory, worded like the real aircraft's caution list, for when a player has the game's picture-in-picture setting disabled — today that silently blanks the FLIR screens and has already produced a false bug report.
- **Cockpit displays:** one master list of which display slot means what (right now three conflicting copies exist and some code uses raw numbers), plus the Apache's tool that auto-generates documentation from that list and fails the build if two things claim the same slot.
- **Hoist:** rewrite its motion loop properly (it currently spins a loop as fast as the computer allows).
- **Performance numbers** (hover power, cruise power, weights) move into reloadable data tables so tuning takes seconds, not rebuilds.

STEP 6 · KEEPING IT GOOD

Documentation & habits

A README that actually explains how the mod is organized; a short contributor guide with the project's rules written down and named (the Chinook team names their rules — "the shadow-not-merge law" — and cites them in reviews, which works); a pull-request checklist; one changelog instead of two; and the Chinook's best habit: every non-obvious decision gets a dated comment explaining why, and failed experiments are kept as labeled warnings so nobody retries them.

Checking the public bug tracker

All 17 open reports on the GitHub project page were reviewed. Good news: every single one fits inside this plan — nothing contradicts it, and several reports independently confirm what the plan already says. Highlights:

- **"No sound dampening in the cockpit"** (confirmed by the team, unfixed) — this is almost certainly the competing-blueprints problem from Step 1. The sound package's settings silently lose the load-order lottery. Step 1 is now done, and this report is the test that proves it worked — it's on the checklist for the group test of the reorganized mod.
- **"Server crashed from network overload with 10 Blackhawks"** — matches two things the audit found: the engine code quietly stacks up duplicate listeners every time someone boards, and several scripts broadcast far more than they need to. Steps 0 and 5 target exactly this.
- **"FLIR tracking worked 50% of the time, now not at all"** — traced and fixed (an update accidentally made the lock key do nothing in the camera view people actually use, and the lock-on search was too fragile to catch big, moving, or flying targets).
- **"Copilot cockpit vanished without anyone changing anything"** — traced and fixed (August 2026). A previous update turned the copilot seat into an "optics turret," which makes the game engine ask the 3-D model for an interior view it doesn't have, so the game fell back to a stripped-down stand-in — hence the see-through console. The copilot seat now uses the proven standard setup on every variant; the dedicated camera-operator seat returns once the model itself gains that missing interior view.
- Two reports turned out to be exact, confirmable bugs — a laser-code script error and dead flight-director keybinds (they ask for a cockpit knob that's spelled differently in the cockpit files). Both went into Step 0. The keybind one also has a second half that lives in the shared Hatchet framework, not this mod — fixed there directly.
- The feature requests are cheaper than they look: the requested IZLID laser exists fully working in the Chinook (copyable), the "add/remove fuel tanks via ACE" request is exactly how the Apache does its equipment swapping (copyable), and the better-waypoints request matches the Apache's navigation package (a design to follow). Each becomes a straightforward port once the groundwork phases are done.

What could go wrong, and the safety nets

- **Step 1 was the riskiest — it moved the most config.** Safety net (used throughout, and it worked): an automated before/after comparison of every helicopter variant's full resolved settings, plus spawning every variant in the editor. Each move landed as its own reviewable change with its comparison attached.
- **Step 5's engine cutover** — keep the old code deletable in a single commit, so if something's wrong we can pinpoint it fast.
- **Multiplayer bugs** — build a small standing test mission (dedicated server + two players, one of whom doesn't own the helicopter). That's exactly how the Chinook team caught their hardest bugs, and it's how the hoist bug here would have been caught.

What a follow-up "legacy audit" turned up (August 20, 2026)

A second, dedicated sweep looked specifically for leftovers from the mod's earlier authorship eras (the project has been through at least three internal naming generations and roughly two dozen contributors). The full list is slotted into the steps above in the technical version; the headlines:

- **A few features are silently broken:** the flight-plan screen's "point the camera at this waypoint" button calls a script that no longer exists, and two other handlers are wired to nothing. Nobody ever reported these — they fail without any message.
- **Developer files ship to every player:** a debug folder (including a captured test-session data dump) and about 48 MB of raw source images are packed into the released mod because the build has no exclusion list. Removing

them makes the download meaningfully smaller.

- **The flight-model package wastes performance:** due to an old registration style, one of its core routines is re-loaded and re-compiled from disk every frame, for every helicopter. Fixing the registration is a free performance win (scheduled with the Step 5 engine work).
- **Credits are missing everywhere:** the authors file has been empty since 2021, almost every package's author field is blank, and the only real credits list is frozen text inside a cockpit display that no longer matches the actual contributor history.
- **Public-facing info has drifted:** the README advertises a version four releases old, the stable changelog is eight releases behind, and two different Discord invites and two different bug-tracker addresses are in circulation — half of visitors are being pointed somewhere stale.

Five decisions for the maintainer

Question	Status / recommendation
Rename the released mod files? (One-time hassle for server owners.)	Decided (Aug 18): the hct branding stays in filenames — target <code>hct_h60_<component></code> with the redundant middle dropped, done during the Step 3 build-tool migration and announced in the changelog so server owners change their lists once. The older internal vtx prefix stays in code for now; renaming it is a known long-term wish but a breaking change that gets its own plan, never a side effect.
Convert all 2,188 old-style names at once, or gradually?	Gradually, as files get touched. A one-shot rewrite is high-risk for zero player benefit.
Fully merge the uh60_config package into the base, or keep it as a small add-on layer?	Decided (Sep 5), and it went further than either option: uh60_config became the single home of <i>all</i> the helicopter's settings, and the big UH60 package now holds only the art (3-D models, textures, sounds). This is the Apache's shape, and it's what shipped with Step 1 — see "How it actually landed" above. It will ship under the name <code>hct_h60_config</code> after the Step 3 renaming.
The hidden Pong game in the MFD package: keep or delete?	Maintainer's call entirely. Keeping it as an easter egg is fine; it just needs its duplicate files cleaned up (already in Step 2).
How far to take translations?	First make English complete in the translation system; invite actual translations later.

Technical companion: docs/IMPROVEMENT_PLAN.md in this folder has every file path, line number, and code pattern behind each item here. This is a plan document — the fixes land through the repository's normal pull requests.